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## SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO1:** Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

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**QUESTIONS:** 4

**Total Marks:**20

### Annexure A SHORT ANSWER TEST

| Criteria  | Conceptual Understanding(3) | Accuracy of Answer(2.5) | Application of Knowledge(2) | Completeness of Response(1.5) | Clarity & Presentation(1) | Total(10) |
|-----------|-----------------------------|-------------------------|-----------------------------|-------------------------------|---------------------------|-----------|
| Weightage | 30%                         | 25%                     | 20%                         | 15%                           | 10%                       | 100%      |
| Q1        |                             |                         |                             |                               |                           |           |
| Q2        |                             |                         |                             |                               |                           |           |
| Q3        |                             |                         |                             |                               |                           |           |
| Q4        |                             |                         |                             |                               |                           |           |

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**Code of Conduct:**

I A. Vinay (19252144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

*Vinay*

## Assessment tool - 1

Evolution from Standalone Computing to cloud-based systems.

\* Cloud Computing evolved through several stages as technology advanced.

### 1. Standalone Computing:-

- \* Each Computer worked independently
- \* Data and Software were stored on the local machine
- \* Sharing files and resources was difficult.

### 2. Client - Server Computing

- \* A central server stored data and applications
- \* Client computers accessed resources from the server



### 3. Internet and Web Technologies:-

- \* The internet enabled users to a CaaS application and information from anywhere
- \* Web browsers made online services easy to use.

### 4. Virtualization

- \* Virtualization allowed multiple virtual machines to run on one physical server
- \* This improved hardware utilization, reduced costs and simplified management.

### 5. Cloud Computing:-

- \* Computing resources such as servers, storage and software are delivered over the internet on demand
- \* Users pay only for the resources they use.

### Key Technological Transitions

- \* standalone computing → client → server architecture
- \* Growth of the internet and web technologies
- \* Development of virtualization
- \* High-speed networking and data centers
- \* Services models such as IaaS, PaaS and SaaS

Example:-

\* A Company that once stored all its files on office Computers can now use cloud storage (such as Google drive or One Drive).

Conclusion:-

\* The evolution from Standalone Computers to cloud Computing has made Computing more flexible, scalable, cost-effective and accessible.

\* A hypervisor is Software that creates and manages Virtual machines (VMs). It allows multiple operating Systems to run on a single Physical Computer by sharing hardware resources such as CPU, memory, storage and network.

Working of a Hypervisor:-

\* The hypervisor is installed on the system

\* It creates multiple virtual machines (VMs)

\* Each VM is allocated CPU, RAM, storage and network resources

\* Every VM runs its own operating system.



# Difference between Type 1 and Type 2 Hypervisors

| Type 1 Hypervisors                                   | Type 2 Hypervisors                                   |
|------------------------------------------------------|------------------------------------------------------|
| * Run directly on the Physical hardware (bare metal) | Runs on top of a host operating system               |
| * Faster and more efficient                          | Slightly slower because it depends on the host OS    |
| * Used in data centers and cloud computing           | used for personal computers, testing and development |
| * More secure and reliable                           | Less secure than Type 1                              |
| Examples: VMware, ESXi, Microsoft, Hyper-V, Xen      | Examples: Oracle, Virtual Box, VMware workstation    |

## Conclusion:-

\* Hypervisors are the foundations of virtualization. Type 1 hypervisors provide high performance for cloud environments, while Type 2 hypervisors are suitable for personal use, software testing and learning.



\* An API is a set of rules that allows different software applications to communicate and exchange data. In cloud computing, APIs enable different cloud services and applications to work together smoothly.

Role of APIs in cloud service integration.

- \* Connect different cloud applications and services
- \* Allow secure data exchange between systems
- \* Enable developers to integrate cloud services into their applications.
- \* Support communication between different cloud platforms.

How APIs support interoperability

- \* APIs use standard protocols such as REST and SOAP, allowing services from different providers to communicate

\* They enable applications running on different platforms

Example :-

\* Google Maps API for location services

\* Payment Gateway API for online Payments

\* Cloud Storage API to store Product images

Conclusion :-

\* APIs are essential in cloud computing because they enable communication, integration, automation and interoperability between different cloud services.



- 28/05/21
- \* Resource Pooling is a cloud Computing feature in which Computing resources such as CPU, memory, storage and network are shared among multiple users.

### How Resource Pooling Works

- \* Cloud Providers maintain a pool of Computing resources
- \* Multiple users share the same Physical infrastructure.
- \* Resources are assigned automatically based on demand

### Benefits of Resource Pooling :-

- \* Efficient utilization of hardware resources
- \* Reduce in infrastructure and maintenance costs
- \* Support multiple users simultaneously



Example: -

\* A cloud provider like AWS or Microsoft Azure uses one data centre to serve thousands of customers. Each customer receives the required storage and computing power.

Conclusion: -

\* Resource pooling allows multiple users to securely share cloud resources, reducing costs and improving ~~efficient~~ efficiency, scalability and resource utilization.

5)

\* Rapid elasticity is a key feature of cloud computing that allows computing that allows computing resources such as CPU, memory, storage and bandwidth to be increased or decreased automatically based on user demand.

low Rapid elasticity support Dynamic workloads

- \* Automatically increases resources when workload increases
- \* Reduces resources when demand decreases
- \* Ensures applications maintain good performance during traffic spikes
- \* Prevents resource wastage and improves efficiency

Benefits :-

- \* Handles sudden increases in user traffic
- \* Improves application performance and availability
- \* Reduces operational costs
- \* Provides flexibility and scalability



Example:-

\* An online Shopping website experiences a large increase in visitors during a festival sale.  
The Cloud Platform automatically adds more servers to handle the extra traffic.

Conclusion:-

\* Rapid elasticity enables cloud systems to quickly adapt to ~~challenge~~ changing workloads by automatically scaling resources up or down. This ensures high performance, cost efficiency, and a better user experience.